TEM characterization of MoS2@SiO2 composites

"How is TEM used to distinguish the MoS2 core from the SiO2 shell" *Generated by LeapSpace*

Direct answer

TEM distinguishes the MoS_2 core from the SiO_2 shell by visualizing a contrast difference between the inner particle and the outer coating: the **MoS_2 core** appears as the denser central region, while the **SiO_2 shell** appears as a surrounding layer with different contrast and measurable thickness [80]. **HRTEM** can add clearer interface detail, making the core–shell boundary easier to see [80].

How it works in practice

- **Core identification:** the MoS_2 region is recognized as the internal, higher-contrast core in the particle image [80].
- **Shell identification:** the SiO_2 layer is seen as the outer coating encasing that core, often allowing the shell thickness to be measured directly [80].
- **Why this is persuasive:** TEM is described as the **direct tool** for verifying core-shell structure, because it shows both morphology and the core–shell spatial arrangement in one image [80].

Best interpretation

If the TEM image shows a **central MoS_2 domain** surrounded by a **continuous outer SiO_2 layer**, that is the clearest evidence of a MoS_2@SiO_2 core-shell structure [80].

Confidence level: High — the conversation explicitly states that TEM/HRTEM directly reveals the dense core, surrounding shell, and shell thickness [80].

References

80. Luo, H.-B., Lu, Z.-J., Qi, X.-D., (...), Yang, J.-H. (2022) Insulating SiO2 internally decorated MoS2 flowers as hybrid filler to simultaneously improve breakdown strength and dielectric performance of PVDF-based composites. *Composites Science and Technology*. https://doi.org/10.1016/j.compscitech.2022.109610